

SQL Subqueries for Sales and Customer Analytics

```
CREATE TABLE Sales (  
SalesID INT AUTO_INCREMENT PRIMARY KEY,  
ProductName VARCHAR(100) NOT NULL,  
Qty INT NOT NULL,  
DateOfPurchase DATE NOT NULL,  
CustomerID INT NOT NULL  
);  
CREATE TABLE Customer (  
CustomerID INT PRIMARY KEY,  
CustomerName VARCHAR(100) NOT NULL,  
PhoneNumber VARCHAR(15)  
);  
INSERT INTO Customer (CustomerID, CustomerName, PhoneNumber) VALUES  
(123, 'Alice', '123-456-7890'),  
(124, 'Bob', '987-654-3210'),  
(125, 'Charlie', '456-789-0123');  
INSERT INTO Sales (ProductName, Qty, DateOfPurchase, CustomerID) VALUES  
( 'Laptop', 10, '2023-01-05', 123),  
( 'Mouse', 50, '2023-01-15', 124),  
( 'Keyboard', 20, '2023-01-20', 123),  
( 'Laptop', 5, '2023-02-10', 125),  
( 'Mouse', 30, '2023-02-25', 123),  
( 'Keyboard', 15, '2023-03-03', 124);  
SELECT  
SalesID,  
ProductName,  
Qty,  
DateOfPurchase,  
(SELECT CustomerName  
FROM Customer c  
WHERE c.CustomerID = s.CustomerID) AS CustomerName  
FROM Sales s;  
SELECT  
c.CustomerID,  
c.CustomerName,
```

```

COALESCE(
(SELECT SUM(Qty)

FROM Sales s
WHERE s.CustomerID = c.CustomerID), 0
) AS TotalSales
FROM Customer c;
SELECT
DISTINCT ProductName,
(SELECT COUNT(DISTINCT CustomerID)
FROM Sales s2
WHERE s2.ProductName = s1.ProductName) AS TotalCustomers
FROM Sales s1;

SELECT
CustomerID,
CustomerName,
PhoneNumber
FROM Customers
WHERE CustomerID NOT IN (SELECT CustomerID FROM Sales);
SELECT ProductName,
(SELECT AVG(Qty)
FROM Sales s2
WHERE s2.ProductName = s1.ProductName) AS AverageQuantity
FROM Sales s1
GROUP BY ProductName;

```

OUTPUT:

Result Grid			
	CustomerID	CustomerName	PhoneNumber
*	NULL	NULL	NULL

Result Grid		
	ProductName	AverageQuantity
▶	Laptop	7.5000
	Mouse	40.0000
	Keyboard	17.5000

